The Bombieri-Stepanov proof of the Hasse-Weil bound

Terence Tao · 2 May 2014

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Let $\mathbf{F}_q$ be a finite field of order $q = p^n$, and let C be an absolutely irreducible smooth projective curve defined over $\mathbf{F}_q$ (and hence over the algebraic closure $k := \overline{\mathbf{F}_q}$ of that field). For instance, C could be the projective elliptic curve

$$C = \{[x,y,z] : y^2 z = x^3 + axz^2 + bz^3\}$$

in the projective plane $\mathbf{P}^2 = \{[x,y,z] : (x,y,z) \neq (0,0,0)\}$, where $a, b \in \mathbf{F}_q$ are coefficients whose discriminant $-16(4a^3 + 27b^2)$ is non-vanishing, which is the projective version of the affine elliptic curve

$$\{(x,y) : y^2 = x^3 + ax + b\}.$$

To each such curve C one can associate a genus g, which we will define later; for instance, elliptic curves have genus 1. We can also count the cardinality $|C(\mathbf{F}_q)|$ of the set $C(\mathbf{F}_q)$ of $\mathbf{F}_q$-points of C. The *Hasse-Weil bound* relates the two:

> **Theorem 1 (Hasse-Weil bound)** $\big||C(\mathbf{F}_q)| - q - 1\big| \leq 2g\sqrt{q}$.

The usual proofs of this bound proceed by first establishing a trace formula of the form

$$|C(\mathbf{F}_{p^n})| = p^n - \sum_{i=1}^{2g} \alpha_i^n + 1 \tag{1}$$

for some complex numbers $\alpha_1, \dots, \alpha_{2g}$ independent of n; this is in fact a special case of the Lefschetz-Grothendieck trace formula, and can be interpreted as an assertion that the zeta function associated to the curve C is rational. The task is then to establish a bound $|\alpha_i| \leq \sqrt{p}$ for all $i = 1, \dots, 2g$; this (or more precisely, the slightly stronger assertion $|\alpha_i| = \sqrt{p}$) is the Riemann hypothesis for such curves. This can be done either by passing to the Jacobian variety of C and using a certain duality available on the cohomology of such varieties, known as Rosati involution; alternatively, one can pass to the product surface $C \times C$ and apply the Riemann-Roch theorem for that surface.

In 1969, Stepanov introduced an elementary method (a version of what is now known as the polynomial method) to count (or at least to upper bound) the quantity $|C(\mathbf{F}_q)|$. The method was initially restricted to hyperelliptic curves, but was soon extended to general curves. In particular, Bombieri used this method to give a short proof of the following weaker version of the Hasse-Weil bound:

> **Theorem 2 (Weak Hasse-Weil bound)** If q is a perfect square, and $q \geq (g+1)^4$, then $|C(\mathbf{F}_q)| \leq q + (2g+1)\sqrt{q} + 1$.

In fact, the bound on $|C(\mathbf{F}_q)|$ can be sharpened a little bit further, as we will soon see.

Theorem 2 is only an upper bound on $|C(\mathbf{F}_q)|$, but there is a Galois-theoretic trick to convert (a slight generalisation of) this upper bound to a matching lower bound, and if one then uses the trace formula (1) (and the "tensor power trick" of sending n to infinity to control the weights α_i) one can then recover the full Hasse-Weil bound. We discuss these steps below the fold.

I've discussed Bombieri's proof of Theorem 2 in this previous post (in the special case of hyperelliptic curves), but now wish to present the full proof, with some minor simplifications from Bombieri's original

presentation; it is mostly elementary, with the deepest fact from algebraic geometry needed being Riemann's inequality (a weak form of the Riemann-Roch theorem).

The first step is to reinterpret $|C(\mathbf{F}_q)|$ as the number of points of intersection between two curves C_1, C_2 in the surface $C \times C$. Indeed, if we define the Frobenius endomorphism Frob_q on any projective space by

$$\mathrm{Frob}_q([x_0, ..., x_n]) := [x_0^q, ..., x_n^q]$$

then this map preserves the curve C, and the fixed points of this map are precisely the $\mathbf{F}_q$ points of C:

$$C(\mathbf{F}_q) = \{z \in C : \mathrm{Frob}_q(z) = z\}.$$

Thus one can interpret $|C(\mathbf{F}_q)|$ as the number of points of intersection between the diagonal curve

$$\{(z, z) : z \in C\}$$

and the Frobenius graph

$$\{(z, \mathrm{Frob}_q(z)) : z \in C\}$$

which are copies of C inside $C \times C$. But we can use the additional hypothesis that q is a perfect square to write this more symmetrically, by taking advantage of the fact that the Frobenius map has a square root

$$\mathrm{Frob}_q = \mathrm{Frob}_{\sqrt{q}}^2$$

with $\mathrm{Frob}_{\sqrt{q}}$ also preserving C. One can then also interpret $|C(\mathbf{F}_q)|$ as the number of points of intersection between the curve

$$C_1 := \{(z, \mathrm{Frob}_{\sqrt{q}}(z)) : z \in C\} \tag{2}$$

and its transpose

$$C_2 := \{(\mathrm{Frob}_{\sqrt{q}}(w), w) : w \in C\}.$$

Let $k(C \times C)$ be the field of rational functions on $C \times C$ (with coefficients in k), and define $k(C_1)$, $k(C_2)$, and $k(C_1 \cap C_2)$ analogously)(although $C_1 \cap C_2$ is likely to be disconnected, so $k(C_1 \cap C_2)$ will just be a ring rather than a field. We then (morally) have the commuting square

$$\begin{array}{ccccc} & & k(C \times C) & & \\ & \swarrow & & \searrow & \\ k(C_1) & & & & k(C_2), \\ & \searrow & & \swarrow & \\ & & k(C_1 \cap C_2) & & \end{array}$$

if we ignore the issue that a rational function on, say, $C \times C$, might blow up on all of C_1 and thus not have a well-defined restriction to C_1. We use $\pi_1 : k(C \times C) \to k(C_1)$ and $\pi_2 : k(C \times C) \to k(C_2)$ to denote the restriction maps. Furthermore, we have obvious isomorphisms $\iota_1 : k(C_1) \to k(C)$, $\iota_2 : k(C_2) \to k(C)$ coming from composing with the graphing maps $z \mapsto (z, \mathrm{Frob}_{\sqrt{q}}(z))$ and $w \mapsto (\mathrm{Frob}_{\sqrt{q}}(w), w)$.

The idea now is to find a rational function $f \in k(C \times C)$ on the surface $C \times C$ of controlled degree which vanishes when restricted to C_1, but is non-vanishing (and not blowing up) when restricted to C_2. On C_2, we thus get a non-zero rational function $f \downharpoonright_{C_2}$ of controlled degree which vanishes on $C_1 \cap C_2$ – which then lets us bound the cardinality of $C_1 \cap C_2$ in terms of the degree of $f \downharpoonright_{C_2}$. (In Bombieri's original argument, one required vanishing to high order on the C_1 side, but in our presentation, we have factored out a $\mathrm{Frob}_{\sqrt{q}}$ term which removes this high order vanishing condition.)

To find this f, we will use linear algebra. Namely, we will locate a finite-dimensional subspace V of $k(C \times C)$ (consisting of certain "controlled degree" rational functions) which projects injectively to $k(C_2)$, but whose projection to $k(C_1)$ has strictly smaller dimension than V itself. The rank-nullity

theorem then forces the existence of a non-zero element P of V whose projection to $k(C_1)$ vanishes, but whose projection to $k(C_2)$ is non-zero.

Now we build V. Pick a $\mathbf{F}_q$ point P_∞ of C, which we will think of as being a point at infinity. (For the purposes of proving Theorem 2, we may clearly assume that $C(\mathbf{F}_q)$ is non-empty.) Thus P_∞ is fixed by Frob_q. To simplify the exposition, we will also assume that P_∞ is fixed by the square root $\mathrm{Frob}_{\sqrt{q}}$ of Frob_q; in the opposite case when $\mathrm{Frob}_{\sqrt{q}}$ has order two when acting on P_∞, the argument is essentially the same, but all references to P_∞ in the second factor of $C \times C$ need to be replaced by $\mathrm{Frob}_{\sqrt{q}} P_\infty$ (we leave the details to the interested reader).

For any natural number n, define R_n to be the set of rational functions $f \in k(C)$ which are allowed to have a pole of order up to n at P_∞, but have no other poles on C; note that as we are assuming C to be smooth, it is unambiguous what a pole is (and what order it will have). (In the fancier language of divisors and Cech cohomology, we have $R_n = H^0(C, \mathcal{O}_C(-nP_\infty))$.) The space R_n is clearly a vector space over k; one can view intuitively as the space of "polynomials" on C of "degree" at most n. When $n = 0$, R_0 consists just of the constant functions. Indeed, if $f \in R_0$, then the image $f(C)$ of f avoids ∞ and so lies in the affine line $k = \mathbf{P}^1 \backslash \{\infty\}$; but as C is projective, the image $f(C)$ needs to be compact (hence closed) in $\mathbf{P}^1$, and must therefore be a point, giving the claim.

For higher $n \geq 1$, we have the easy relations

$$\dim(R_{n-1}) \leq \dim(R_n) \leq \dim(R_{n-1}) + 1. \tag{3}$$

The former inequality just comes from the trivial inclusion $R_{n-1} \subset R_n$. For the latter, observe that if two functions f, g lie in R_n, so that they each have a pole of order at most n at P_∞, then some linear combination of these functions must have a pole of order at most $n-1$ at P_∞; thus R_{n-1} has codimension at most one in R_n, giving the claim.

From (3) and induction we see that each of the R_n are finite dimensional, with the trivial upper bound

$$\dim(R_n) \leq n + 1. \tag{4}$$

Riemann's inequality complements this with the lower bound

$$\dim(R_n) \geq n + 1 - g, \tag{5}$$

thus one has $\dim(R_n) = \dim(R_{n-1}) + 1$ for all but at most g exceptions (in fact, exactly g exceptions as it turns out). This is a consequence of the Riemann-Roch theorem; it can be proven from abstract nonsense (the snake lemma) if one defines the genus g in a non-standard fashion (as the dimension of the first Cech cohomology $H^1(C)$ of the structure sheaf $\mathcal{O}_C$ of C), but to obtain this inequality with a standard definition of g (e.g. as the dimension of the zeroth Cech cohomolgy $H^0(C, \Omega^1_C)$ of the line bundle of differentials) requires the more non-trivial tool of Serre duality.

At any rate, now that we have these vector spaces R_n, we will define $V \subset k(C \times C)$ to be a tensor product space

$$V = R_\ell \otimes R_m$$

for some natural numbers $\ell, m \geq 0$ which we will optimise in later. That is to say, V is spanned by functions of the form $(z, w) \mapsto f(z)g(w)$ with $f \in R_\ell$ and $g \in R_m$. This is clearly a linear subspace of $k(C \times C)$ of dimension $\dim(R_\ell)\ \dim(R_m)$, and hence by Rieman's inequality we have

$$\dim(V) \geq (\ell + 1 - g)(m + 1 - g) \tag{6}$$

if

$$\ell, m \geq g - 1. \tag{7}$$

Observe that $\iota_1 \circ \pi_1$ maps a tensor product $(z, w) \mapsto f(z)g(w)$ to a function $z \mapsto f(z)g(\mathrm{Frob}_{\sqrt{q}} z)$. If $f \in R_\ell$ and $g \in R_m$, then we see that the function $z \mapsto f(z)g(\mathrm{Frob}_{\sqrt{q}} z)$ has a pole of order at most $\ell + m\sqrt{q}$ at P_∞. We conclude that

$$\iota_1 \circ \pi_1(V) \subset R_{\ell+m\sqrt{q}} \tag{8}$$

and in particular by (4)

$$\dim(\pi_1(V)) \leq \ell + m\sqrt{q} + 1 \tag{9}$$

and similarly

$$\dim(\pi_2(V)) \leq \ell\sqrt{q} + m + 1. \tag{10}$$

We will choose m to be a bit bigger than ℓ, to make the π_2 image of V smaller than that of π_1. From (6), (10) we see that if we have the inequality

$$(\ell + 1 - g)(m + 1 - g) > \ell\sqrt{q} + m + 1 \tag{11}$$

(together with (7)) then π_2 cannot be injective.

On the other hand, we have the following basic fact:

> **Lemma 3 (Injectivity)** If
>
> $$\ell < \sqrt{q}, \tag{12}$$
>
> then $\pi_1 : V \to \pi_1(V)$ is injective.

Proof: From (3), we can find a linear basis $f_1, \ldots, f_a$ of R_ℓ such that each of the f_i has a distinct order d_i of pole at P_∞ (somewhere between 0 and ℓ inclusive). Similarly, we may find a linear basis $g_1, \ldots, g_b$ of R_m such that each of the g_j has a distinct order e_j of pole at P_∞ (somewhere between 0 and m inclusive). The functions $z \mapsto f_i(z)g_j(\mathrm{Frob}_{\sqrt{q}} z)$ then span $\iota_1(\pi_1(V))$, and the order of pole at P_∞ is $d_i + \sqrt{q}e_j$. But since $\ell < \sqrt{q}$, these orders are all distinct, and so these functions must be linearly independent. The claim follows. □

This gives us the following bound:

> **Proposition 4** Let ℓ, m be natural numbers such that (7), (11), (12) hold. Then $\left|C(\mathbf{F}_q)\right| \leq \ell + m\sqrt{q}$.

Proof: As π_2 is not injective, we can find $f \in V$ with $\pi_2(f)$ vanishing. By the above lemma, the function $\iota_1(\pi_1(f))$ is then non-zero, but it must also vanish on $\iota_1(C_1 \cap C_2)$, which has cardinality $\left|C(\mathbf{F}_q)\right|$. On the other hand, by (8), $\iota_1(\pi_1(f))$ has a pole of order at most $\ell + m\sqrt{q}$ at P_∞ and no other poles. Since the number of poles and zeroes of a rational function on a projective curve must add up to zero, the claim follows. □

If $q \geq (g+1)^4$, we may make the explicit choice

$$m := \sqrt{q} + 2g; \quad \ell := \left\lfloor \frac{g}{g+1}\sqrt{q} \right\rfloor + g + 1$$

and a brief calculation then gives Theorem 2. In some cases one can optimise things a bit further. For instance, in the genus zero case $g = 0$ (e.g. if C is just the projective line $\mathbf{P}^1$) one may take $\ell = 1, m = \sqrt{q}$ and conclude the absolutely sharp bound $\left|C(\mathbf{F}_q)\right| \leq q + 1$ in this case; in the case of the projective line $\mathbf{P}^1$, the function f is in fact the very concrete function $f(z, w) := z - w^{\sqrt{q}}$.

> **Remark 1** When $q = p^{2n+1}$ is not a perfect square, one can try to run the above argument using the factorisation $\mathrm{Frob}_q = \mathrm{Frob}_{p^n}\mathrm{Frob}_{p^{n+1}}$ instead of $\mathrm{Frob}_q = \mathrm{Frob}_{\sqrt{q}}\mathrm{Frob}_{\sqrt{q}}$. This gives a weaker version of the above bound, of the shape $\left|C(\mathbf{F}_q)\right| \leq q + O(\sqrt{p}\sqrt{q})$. In the hyperelliptic case at least, one can erase this loss by working with a variant of the argument in which one requires f to vanish to high order at C_1, rather than just to first order; see this survey article of mine for details.

— 1. Additional notes —

One can get a "cheap" proof of Riemann's inequality (with the "wrong" definition of the genus g) from Cech cohomology as follows. For any natural number n, let $\mathcal{O}_C(-nP_\infty)$ denote the sheaf on C, defined

by setting the sections on any open set U of C to be the rational functions with no poles at U, except possibly for a pole of order up to n at P_∞ in the case that U contains P_∞. Then R_n is the space of global sections on this sheaf, that is to say the zeroth Cech cohomology $H^0(\mathcal{O}_C(-nP_\infty))$. On the other hand, we also have the skyscraper sheaf k_{P_∞} on C, defined by setting the sections on U to be k if U contains P_∞ and $\{0\}$ otherwise. For $n \geq 1$, we have an obvious short exact sequence of sheaves

$$0 \to \mathcal{O}_C(-(n-1)P_\infty) \to \mathcal{O}_C(-nP_\infty) \to k_{P_\infty} \to 0$$

which upon taking Cech cohomology (and using the snake lemma) gives the long exact sequence

$$0 \to H^0(\mathcal{O}_C(-(n-1)P_\infty)) \to H^0(\mathcal{O}_C(-nP_\infty)) \to H^0(k_{P_\infty}) \to$$

$$H^1(\mathcal{O}_C(-(n-1)P_\infty)) \to H^1(\mathcal{O}_C(-nP_\infty)) \to H^1(k_{P_\infty}).$$

One can compute that $H^0(k_{P_\infty}) = k$ and $H^1(k_{P_\infty}) = 0$, so the long exact sequence becomes

$$0 \to R_{n-1} \to R_n \to k \to$$

$$H^1(\mathcal{O}_C(-(n-1)P_\infty)) \to H^1(\mathcal{O}_C(-nP_\infty)) \to 0.$$

The first part of this long exact sequence already recovers the trivial bounds (3). But it also shows that the dimensions of the spaces $H^1(\mathcal{O}_C(-nP_\infty))$ are non-increasing, and decrease by one precisely when the left inequality in (3) is sharp. If one then makes the non-standard definition $g := \dim(H^1(\mathcal{O}_C))$, then this decrease can occur at most g times, and Riemann's inequality then follows. (However, it is not immediately obvious with this definition that g is finite; this requires some additional effort, e.g. invoking the Riemann-Hurwitz formula.) To relate this definition of g to the more usual notions of genus requires the use of Serre duality, which will not be discussed here.

The upper bound in Theorem 2 (or more precisely, a generalisation of this bound) can be converted into a comparable lower bound, namely

> **Theorem 5** Let C be an absolutely irreducible quasiprojective curve of bounded degree, defined over $\mathbf{F}_q$, and let $q = p^n$ be a perfect square. Then
>
> $$\big|C(\mathbf{F}_q)\big| = q + O(\sqrt{q}),$$
>
> where the implied constant can depend on C but is uniform in n in the limit $n \to \infty$ (holding p and C fixed).

Proof: The upper bound follows from Theorem 2, after first removing all the singularities from C, normalising C to be projective, and noting that the genus does not depend on n. (Note that removing singularities only deletes $O(1)$ points at most.) So the remaining task is to establish the matching lower bound. Unfortunately, the trace formula (1) is not enough by itself to convert upper bounds to lower bounds, due to the case in which all the α_i are positive, although strangely enough it can be used for the reverse task of converting lower bounds to upper bounds. Instead, we use a Galois-theoretic argument, though for (somewhat idiosyncratic) reasons, I will disguise the Galois theory by writing everything in a geometric language rather than an algebraic one.

The basic idea is to embed C (perhaps with a bounded number of points removed) as a component of a larger (non-reducible) curve C' for which (a) the number of "rational points" on this larger curve C' can be counted almost exactly; and (b) an upper bound similar to Theorem 2 exists for the number of "rational points" on each of the components of C', so that a lower bound can then be established by subtraction.

An easy instance of this trick arises in the case of hyperelliptic curves, which we will write affinely (deleting the point at infinity, by abuse of notation) as $C = \{(x,y) : y^2 = f(x)\}$ for some polynomial f defined over $\mathbf{F}_q$ that is not a perfect square. We can embed this in the reducible curve

$$C' := \{(x,y) : y^2 = f(x)\} \cup \{(x,y) : y^2 = af(x)\}$$

where a is an arbitrarily chosen quadratic non-residue of $\mathbf{F}_q$. The curve C' is the union of two curves, one of which is a dilate of the other, so they both have the same genus g, and so they each have at most $q + O(\sqrt{q})$ points (in the regime when g is bounded and q is a perfect square), thanks to Theorem 2. On the other hand, if $f(x)$ is non-zero, then exactly one of $f(x)$ and $af(x)$ is a quadratic residue, and so for all but a bounded number of x, there are exactly two values of y with $(x,y) \in C'$, and so $\left|C'(\mathbf{F}_q)\right| = 2q + O(1)$. One can then subtract the upper bound for one of the components from this estimate to obtain a matching lower bound for the other component, and in particular $\left|C(\mathbf{F}_q)\right| \geq q - O(\sqrt{q})$.

In general we can proceed as follows. Let d be the degree of C, so $d = O(1)$. After deleting $O(1)$ points from C, we can view C as a degree d cover of the affine line $\mathbf{A}^1$ with $O(1)$ many points removed, with projection map $\pi : C \to \mathbf{A}^1$. Once we do so, we can pass to the lifted curve $\tilde{C} := \bigwedge^d C$, defined as the collection of distinct d-tuples $y_1, \ldots, y_d$ in C that lie over the same point in $\mathbf{A}^1$, thus $\pi(y_1) = \ldots = \pi(y_d)$. This is a degree $d!$ cover of (most of) $\mathbf{A}^1$, and is defined over $\mathbf{F}_q$. It also carries a free action of the permutation group S_d: if $\sigma \in S_d$, we define

$$\sigma(y_1, \ldots, y_d) := (y_{\sigma^{-1}(1)}, \ldots, y_{\sigma^{-1}(d)})$$

(the inverse is there to make the action a left action; one could also work with right actions instead if desired).

The curve $\tilde{C}$ need not be absolutely irreducible, so we break it into absolutely irreducible components. The permutation group S_d acts transitively on these components, so if we pick one of these components, say $\tilde{C}_0$, and let $H := \{\sigma \in S_d : \sigma\tilde{C}_0 = \tilde{C}_0\}$ be the stabiliser, then H is a subgroup of S_d (in particular, $H = O(1)$), and one can index the components of $\tilde{C}_0$ as $\sigma\tilde{C}_0$, as σ ranges over a set of coset representatives of S_d/H. Each fibre of $\tilde{C}_0$ over a generic point of the base $\mathbf{A}^1$ is an orbit of H; since the map $(y_1, \ldots, y_d) \to y_1$ projects $\tilde{C}_0$ to the connected curve C, we conclude that the projection of this orbit must have cardinality d. In other words, the permutation group H is a transitive group.

Even though $\tilde{C}$ is defined over $\mathbf{F}_q$, the individual components $\sigma\tilde{C}_0$ of $\tilde{C}$ need not be. In particular, the Frobenius image $\mathrm{Frob}_q\tilde{C}_0$ of $\tilde{C}_0$ might be another component of $\tilde{C}_0$, say $\rho_q\tilde{C}_0$.

Now let x be a generic point of $\mathbf{A}^1$ defined over $\mathbf{F}_q$, then the fibre of $\tilde{C}_0$ above x is a free H-orbit and thus has cardinality $|H|$ orbit. Let y be a point in this fibre. Applying Frobenius, we see that $\mathrm{Frob}_q(y)$ lies in the fibre of $\rho_q\tilde{C}_0$ above x, and hence

$$\mathrm{Frob}_q(y) = \rho_q\sigma y \tag{13}$$

for a unique $\sigma \in H$. Of course, if $y \in \tilde{C}_0$ does not lie over a point x defined over $\mathbf{F}_q$, then the equation (13) cannot hold. Since there are $q - O(1)$ points in $\mathbf{A}^1$ with $O(1)$ points removed that are defined over $\mathbf{F}_q$, and each of these give rise to $|H|$ points in $\tilde{C}_0$, we conclude that

$$\sum_{\sigma\in H}\left|\{y \in \tilde{C}_0 : \mathrm{Frob}_q(y) = \rho_q\sigma y\}\right| = q|H| - O(1). \tag{14}$$

On the other hand, a modification of Theorem 2 shows the upper bound

$$\left|\{y \in \tilde{C}_0 : \mathrm{Frob}_q(y) = \rho_q\sigma y\}\right| \leq q + O(\sqrt{q}) \tag{15}$$

for each $\sigma \in H$. Strictly speaking, Theorem 2 is only directly applicable in the case when $\rho_q\sigma$ is the identity. However, even when $\rho_q\sigma$ is not the identity, one can "twist" the proof of Theorem 2 to establish the above estimate, basically by replacing the curve C_1 in (2) by a twisted variant

$$C_1' := \{(\rho_q\sigma z, \mathrm{Frob}_{\sqrt{q}}(z)) : z \in C\}$$

which is still isomorphic to C_1; we leave the details to the interested reader. If we subtract $|H| - 1$ instances of (15) from (14), we obtain that

$$\left|\{y \in \tilde{C}_0 : \mathrm{Frob}_q(y) = \rho_q\sigma y\}\right| \geq q - O(\sqrt{q})$$

for each $\sigma \in H$, which on combining with (15) gives

$$\left|\{y \in \tilde{C}_0 : \mathrm{Frob}_q(y) = \rho_q \sigma y\}\right| = q + O(\sqrt{q})$$

for each $\sigma \in H$.

Next, as H acts transtively on $\{1, ..., d\}$, we see that there are $|H|/d$ values of σ such that $\rho_q \sigma(1) = 1$. We conclude that

$$\left|\{(y_1, ..., y_d) \in \tilde{C}_0 : \mathrm{Frob}_q(y_1) = y_1\}\right| = \frac{|H|}{d} q + O(\sqrt{q}).$$

But as C has degree d above $\mathbf{A}^1$, and $\tilde{C}_0$ has degree $|H|$, we see that for generic y_1 there are exactly $|H|/d$ points $(y_1, ..., y_d)$ in $\tilde{C}_0$ that lie above y_1. We conclude that

$$\left|\{y_1 \in C : \mathrm{Frob}_q(y_1) = y_1\}\right| = q + O(\sqrt{q}),$$

and the claim follows. □

Finally, if one uses the trace formula (1), then Theorem 5 implies the bounds

$$|\alpha_i| \leq \sqrt{p} \tag{16}$$

for all $i = 1, ..., 2g$, and hence Theorem 1. Indeed, if we let α_i be of maximal magnitude, thus $|\alpha_i| \geq |\alpha_j|$ for all $j = 1, ..., 2g$, and an easy application of the Dirichlet approximation theorem then shows that

$$\left|\sum_{j=1}^{2g} \alpha_j^n\right| \geq |\alpha_i|^n$$

for infinitely many even n, which when combined with Theorem 5 implies that

$$|\alpha_i|^n = O(p^{n/2})$$

for infinitely many even n. Taking n^{th} roots and then sending n to infinity (cf. the "tensor power trick") we obtain (16) as required.

Comments (18)

Che! · 2 May 2014

I think you have typo “elliptic curves have genus 1”
[Corrected, thanks – T.]

> **Anonymous** · 9 May 2014
>
> I still see “elliptic curves have genus 1” in the text. Why is it a typo?
> *[This is the corrected form of the text; previously I had written “elliptic curves had genus g“. -T.]*

Andrew S · 2 May 2014

I think there is a small typo near the beginning. Is the number of rational points $C[\mathbb{F}_q]$ or $C[\mathbb{F}_q]|$ or $|C(\mathbb{F}_q)|$? A nice post, nonetheless. Though, if you think the definition $g = H^1(\mathcal{O}_C)$ is wrong, then I wonder what you must think of the definition via the Hilbert polynomial!
[Corrected, thanks – T.]

Lior Silberman · 3 May 2014

There’s an arrow missing in the second version of the long exact sequence in cohomology (after the k). Also, I think “non-reducible” was supposed to be “non-irreducible”.
[Corrected, thanks – T.]

> **Lior Silberman** · 3 May 2014
>
> Also, in the formula after (16) you use i both as a bound and as a free variable, with the value of the free instance fixed by the previous sentence.
> *[Corrected, thanks – T.]*

Brian Mingus · 3 May 2014

It would be awesome if you could include a note indicating why a particular proof is interesting and important or what about it is novel for those of us more lay readers who still enjoy following your technical blog.

O Ahmadi · 15 May 2014

There is a typo in the formula for the elliptic discriminant. The coefficient of b^2 should be 27.
[Corrected, thanks – T.]

Angel Fernandez · 5 July 2014

It is true the projection in some of the elliptic curves are integers. He has demonstrated the mathematical Enfer Diez; by an equation, we can deternine whether there is an elliptical curve.
See in: Universal Journal of Computational Mathematics Vol 1 (1);2013
Title: Equation for solve elliptical curves.

משפט הסה בהסתברות גבוהה (מיני-פוסט) | One and One · 14 August 2015

[…] עם תכונות התאפסות מסויימות, ואפשר לקרוא עליה כאן או כאן. למעשה, ההוכחה שלו לא הייתה חזקה מספיק כדי לתת את הקבוע […]

A function field analogue of Riemann zeta statistics | What's new · 17 May 2019

[…] can in turn be established by elementary methods such as Stepanov’s method, as discussed in this previous post). The non-squarefree case is more complicated (and can be established using the machinery of […]

Recent Progress on a Problem of Sárközy | George Shakan · 28 May 2019

[...] now proceed to the proof of Theorem 1, which adopts the Stepanov method of auxiliary [...]

Anonymous · 18 April 2022

Legend!!

Anonymous · 29 November 2023

Hello,
There are two details of this proof I don't quite follow:
1. Does Frob_sqrt{q} preserve C? If so, why?
Possible counter-example: If q = 25 and C is the elliptic curve y^{2} = x^{3} + aX where a is an element of order 24 in the finite field F_{25}, the point [1:(1+a)^{1/2}:1] is mapped to [1:(1+a)^{5/2}:1], which does not appear to satisfy the original curve?
2. I think Equation 11 is too strong of an inequality for the choice of l and m following Proposition 4:
Possible counter-example: If q = 25 and g = 1, then l = 4 and m = 7, and both sides of Equation 11 are equal to 28.
Note that Equation 11 can be weakened to fix this problem: As in Bombieri's original proof, as long as l and m are at least g, m + l(sqrt(q)) is at least 2g. By Riemann-Roch, the dimension of R_{m+l(sqrt(q))} is then equal to m + l(sqrt(q))+1-g.
Thank you for your assistance.

> **Terence Tao** · 2 December 2023
>
> Thanks for the corrections. For issue 1, the intent was to assume that C was in fact defined over $\mathbf{F}_{\sqrt{q}}$ and not just $\mathbf{F}_q$. For issue 2, I guess my intended simplification of just using Riemann's inequality instead of Riemann-Roch does not quite work. The simplest thing to do is indeed just to point back to Bombieri's original paper for a full treatment.
>
> > **Anonymous** · 3 December 2023
> >
> > That makes sense, and considering curves defined over F_sqrt(q) at first is sufficient to prove the Hasse-Weil bound because Theorem 5 concerns the asymptotic behavior of the number of points on the curve, so there is no harm in making a finite base field extension. Thank you.

Anonymous · 12 August 2024

Could you elaborate on the "easy application of Dirichlet's theorem"? All the sources for this proof suggest it is obvious but I can't for the life of me figure it out.

> **Terence Tao** · 16 August 2024
>
> Write α_j in polar coordinates as $\alpha_j = r_j e^{2\pi\theta_j}$, thus $\alpha_j^n = r_j^n e^{2\pi i n\theta_j}$. Then use the (simultaneous version of the) Dirichlet approximation theorem to find n for which all of the $n\theta_j$ are close to an integer.
>
> > **Anonymous** · 16 August 2024
> >
> > I see! Thanks prof., I was going insane trying to see what to apply the theorem to.